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STAT 437

LEC 801

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Assignment 3

Q1

a)

Coefficient	**Estimate**	**P-Value**
Intercept	-3.37395	<2e-16
Age	-0.17676	0.0093
Smoke	0.41478	0.1484

Fitted Model : $\log(\mu_{ij} / 1-\mu_{ij}) = (-3.37395 + r_i) - 0.17676(age_{ij}) + 0.41478(smoke_{ij})$

Estimated standard deviation of random intercept : $\sigma_r = 2.343$

b)

From the odds-ratio of age = $e^{-0.17676} \approx 0.838$, for each one-year increase in age, the estimated odds of wheezing are multiplied by a factor of 0.838. That is, for an additional year of age, holding smoke status and child's random intercept constant, the estimated odds decrease by approximately 16.2% ((1-0.838) x 100%).

The corresponding odds-ratio for the estimated smoking coefficient $e^{0.41478.} \approx 1.514$. This suggests that maternal smoking has approximately 51.4% higher estimated odds of wheezing compared with mothers who have not smoked, holding age and random effects constant. Though, with a p-value 0.1484 > 0.05 there is insufficient evidence to conclude that maternal smoking is associated with the odds of wheezing at a 5% significance level.

c)

Random intercepts account for underlying correlation of data between the same individuals across repeated observations, in this context the random intercept accounts for the baseline differences between children's proclivity to wheezing. The estimated standard deviation of the random intercept is 2.343, this represents this variability in the baseline log-odds of wheezing

among the children. This suggests substantial differences in the underlying likelihood of wheezing among the children. That is children with positive random intercept values have higher baseline odds probabilities while those with negative values have lower baseline odds and probabilities.

Q2

a)

Fixed Effects

Coefficient	**Estimate**	**Standard Error**	**P-Value**
Intercept	-3.38218	0.32655	<0.001
Age	-0.17678	0.06797	0.0093
Smoke	0.43360	0.48936	0.3756

Fitted Model : $\log(\mu_{ij} / 1-\mu_{ij}) = (-3.38218 + r_{0i}) - 0.17678(age_{ij}) + (0.43360 + r_{1i})(smoke_{ij})$

Random Effects

Random Effect	**Estimated variance**	**Estimated Standard Deviations**
Random Intercept	5.5346	2.3526
Random smoking effect	0.2257	0.4751

b)

Age:

Odds-ratio: $e^{-0.17678} \approx 0.838$ corresponds to 16.2% ((1-0.838) x 100%)

So holding smoking status and individual specific random effects at constant, each additional year age is associated with approximately 16.2% lower estimated odds of wheezing. With a p-value of 0.0093 this indicates a statistically significant inference at 5% level of significance. However the model failed to converge. Model 2 produced convergence warnings, specifically citing inability to evaluate the scaled gradient. Further examination found that the maternal

smoking status remained constant with all 537 children. As such, the random intercept and random smoking effect cannot be identified reliably so this result should not be treated as a conclusion.

```
optimizer (Nelder_Mead) convergence code: 0 (OK)
unable to evaluate scaled gradient
Model failed to converge: degenerate  Hessian with 1 negative eigenvalues
  See ?lme4::convergence and ?lme4::troubleshooting.
> # checking for a singular fit
> isSingular(model2)
[1] FALSE
> # check convergence warnings
> model2@optinfo$conv$lme4$messages
[1] "unable to evaluate scaled gradient"
[2] "Model failed to converge: degenerate  Hessian with 1 negative
eigenvalues\n  See ?lme4::convergence and ?lme4::troubleshooting."
```

Smoking:

Odds-ratio: $e^{0.43360.} \approx 1.543$ corresponds to 54.3% ((1.543 - 1) x 100%)

The estimated smoking coefficient is 0.43360 corresponding to the odds ratio of 1.543. Thus, holding age constant, the random-effect values smoking is associated with approximately 54.3% higher estimated odds of wheezing compared to no maternal smoking. Though, p-value 0.3756 > 0.05, as such, this should not be considered statistically significant according to the test.

Compared to Question 1, the first model assumes the same smoking coefficient across the children, while model 2 allows the smoking coefficient to vary.

c)

Estimated standard deviation of the random effect associated with smoke:

$$\sigma_{r1} = 0.4751$$

This describes how much the child specific smoking effects vary around the smoking coefficient. A large σ_{r1} would indicate great variability in smoking effects across the children. However, as smoking status does not vary between each child and since the model failed to converge, this result cannot be reliably interpreted.

d)

The estimated correlation between random intercept and random smoking effect is -0.15. This negative correlation suggests that children with a higher baseline wheezing susceptibility tend to have smaller smoking related effects, while a positive correlation would suggest children with higher baseline susceptibility tend to have larger smoking effects. Again, since the model failed to converge, this result cannot be reliably interpreted.

Q3

a)

In a regular LMM the response is assumed to be of continuous value with normally distributed errors, as such the expected response can be modeled as a linear combination of fixed and random effects. In a GLMM, the response can have different distributions depending on the type of response values suitable for the problem such as binary or count responses. In this situation, we use a link function to control the expected response to the appropriate distribution. For instance, logit link for binary response or log link for count responses for Poisson GLMMs.

b)

Random effects are introduced in LMEs to account for longitudinal data, that is, when repeated measurements of the same individuals may be correlated because of underlying conditions. For example if studying blood pressure of same individuals across time, individuals with underlying conditions may be disproportionately represented in the data. Fixed effects represent overall population relationships between the predictors and response, random effects represent individual deviations from the population relationships. The random intercept represents the baseline response value for each individual which can be different from the population. For example, if studying a student's test scores across time, a student that generally performs higher than the population average will have a higher intercept value.

c)

The common approach to estimate parameters of a LMM is using Maximum Likelihood (ML) or Restricted Maximum Likelihood and Expectation Maximization (EM) or Newton-Raphson may be used during iteration. The main approaches to estimating parameters of a GLMM are Marginal Likelihood approximated using Monte Carlo EM and Numerical Integration methods. It is much harder to estimate the parameters of GLMM models as the values for random effects are not known ahead of time as such we must consider the different values that they could take on weighted by their probabilities. The likelihood calculations require integration over all possible values and these solutions often have multi-dimensional integrals with non-closed-form solutions.

d)

A standard logistic regression model may not be appropriate in this situation as the model assumes that the observations are independent, however, repeated measurements of the same individual across time may be correlated due to underlying individual characteristics. As such, if correlations are ignored this can lead to different conclusions about statistical significance, unreliable standard errors and confidence intervals. A logistic GLM addresses this issue by introducing random effects such as random intercept, allowing each individual to have their own baseline log-odds of the response. GLMM assumes that after this conditioning on random effects and predictors then the repeated binary responses are independent.

Appendix

```
# Q1
library(faraway)
library(lme4)

data("ohio")
head(ohio)
str(ohio)
summary(ohio)

# resp: Indicator of wheeze status (1 = yes, 0 = no).
# id: Subject identification number.
# age: Age of the child, where age = 0 corresponds to 9 years old.
# smoke: Indicator of maternal smoking at the first year of the study

# Q1 a)
model1 <- glmer(
  resp ~ age + smoke + (1 | id),
  family = binomial(link = "logit"),
  data = ohio
)

summary(model1)

# Q2
# random intercept and random smoking effect
model2 <- glmer(
  resp ~ age + smoke + (1 + smoke | id),
```

```
  family = binomial(link = "logit"),
  data = ohio
)

summary(model2)

# checking for a singular fit
isSingular(model2)

# examine estimated random effects
VarCorr(model2)

# check convergence warnings
model2@optinfo$conv$lme4$messages

# whether smoking status varies within children
smoke_variation <- tapply(
  ohio$smoke,
  ohio$id,
  function(x) length(unique(x))
)

table(smoke_variation)
```